

# Ali Raza

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## SUMMARY

AI/ML Engineer with 3+ years building and deploying production ML systems end-to-end — computer vision pipelines, inference optimization, and full-stack ML products. Core expertise across PyTorch, ONNX/TensorRT, FastAPI, Docker, and cloud MLOps stacks (MLflow, Prometheus/Grafana, GitHub Actions). Comfortable owning the full lifecycle: training, benchmarking, backend service architecture, and cost-aware production deployment.

## TECHNICAL SKILLS

**ML/DL:** PyTorch, TensorFlow, Keras, Scikit-learn, HuggingFace, XGBoost, LightGBM, Transfer Learning, Fine-tuning, LoRA, QLoRA, PEFT

**Computer Vision:** YOLOv8, U-Net, TransUNet, ViT, SAM, ResNet, DenseNet, BioViL-T, OpenCV, GradCAM, CBAM

**LLMs & NLP:** LangChain, LangGraph, LlamaIndex, RAG, FAISS, Pinecone, BERT, Embeddings, Prompt Engineering, OpenAI API, Gemini API

**MLOps:** MLflow, Docker, Kubernetes, Kubeflow, FastAPI, Prometheus, Grafana, GitHub Actions, CI/CD, AWS EC2

**Programming:** Python, C++, JavaScript, Pandas, NumPy, SQL, PostgreSQL, MongoDB, Git

## EXPERIENCE

**ML Engineer — Freelance** | International Clients 2023 – Present  
Remote, Faisalabad, Pakistan

- Delivered **end-to-end AI/ML solutions** for multiple freelance clients, from data preparation and model development to scalable production deployment.
- Specialized in **computer vision, LLM applications, and MLOps**, building production-ready systems tailored to client requirements.
- Worked with technologies including **YOLOv8, FastAPI, Docker, HuggingFace, LangChain, FAISS, Pinecone, MLflow, and GitHub Actions**.
- Deployed and maintained AI services on cloud GPU infrastructure with emphasis on scalability, automation, and reliable inference.

**Data Analyst Intern** | TAKMIL 2024  
Louisville, Kentucky, USA · Remote

- Cleaned and analyzed large-scale datasets via Python (Pandas, NumPy, SQL); built **Power BI dashboards** — proposed optimizations improving workflow efficiency by **30%**.

## RESEARCH & KEY PROJECTS

**RadGuard: Multimodal ML System for Medical Imaging** — Final Year Thesis 2025–2026  
*BioViL-T · CXR-BERT · PyTorch · MLflow · Docker · Kubernetes · Kubeflow · FastAPI · AWS EC2*

- Built a multimodal deep learning system to detect 4 error types in AI-generated chest X-ray reports across 14 conditions using 74K+ MIMIC-CXR samples, achieving **0.66 validation F1**.
- Designed a production-ready MLOps pipeline with **Docker, Kubernetes, Kubeflow, MLflow, FastAPI, and AWS EC2**, enabling scalable deployment, automated retraining, monitoring, and FAISS-powered RAG querying.

**AI Cost Segregation Platform** — Production SaaS 2025–Present  
*YOLOE · SAM2 · Gemini VLM · FastAPI · Firebase · Cloudinary · ReportLab*

- Technical lead on a full-stack AI SaaS turning property photos into IRS-defensible tax depreciation reports.
- Redesigned a failing single-model classifier into a **3-stage YOLOE + SAM2 + Gemini VLM pipeline** against a 312-item taxonomy.
- Rearchitected deployment from Vast.ai GPU rental to **CPU-only VPS + hosted VLM inference**; owned MLOps roadmap (Docker, GitHub Actions CI/CD, MLflow, HITL correction loop).

**RetailSense AI: Production ML Forecasting Platform** 2025  
*LightGBM · XGBoost · ARIMA · K-Means · MLflow · FastAPI · Docker · Kubernetes · AWS EC2 · React*

- Built a retail demand forecasting platform for a 54-store grocery chain, where LightGBM outperformed XGBoost and ARIMA, achieving **RMSE 336 and MAPE 8.78% (55% improvement)** over baseline).
- Implemented a production MLOps pipeline using **MLflow, FastAPI, Docker, Kubernetes, and AWS EC2**, with SHAP explainability, KS-test drift monitoring, and statistical model validation.

**InferBench: ML Inference Optimization & Serving Platform** 2026  
*ViT-Base · ONNX Runtime · TensorRT (FP16) · FastAPI · Redis · PostgreSQL · Prometheus · Grafana*

- Optimized ViT-Base inference using ONNX Runtime and TensorRT (FP16), achieving **2.9×–5.6× lower latency** while preserving **100% prediction consistency**.
- Developed a production inference platform featuring **custom dynamic batching**, Redis caching (**2377× faster repeat requests**), monitoring with Prometheus/Grafana, and Docker-based CI/CD.

## EDUCATION

**BSc Computer Science** | **FAST-NUCES, Faisalabad** 2022 – 2026  
Co-Head Hackathon ACM Daira 2025 | TAKMIL Data Science Internship 2024

## PUBLICATIONS

Ali Raza et al. “Segmentation & Classification of CIEDs in Chest X-rays” — **ACR-2026**